

Module 2

SENSORS & CONNECTIVITY

Sensing

Generally, sensors are used in the architecture of IOT devices.

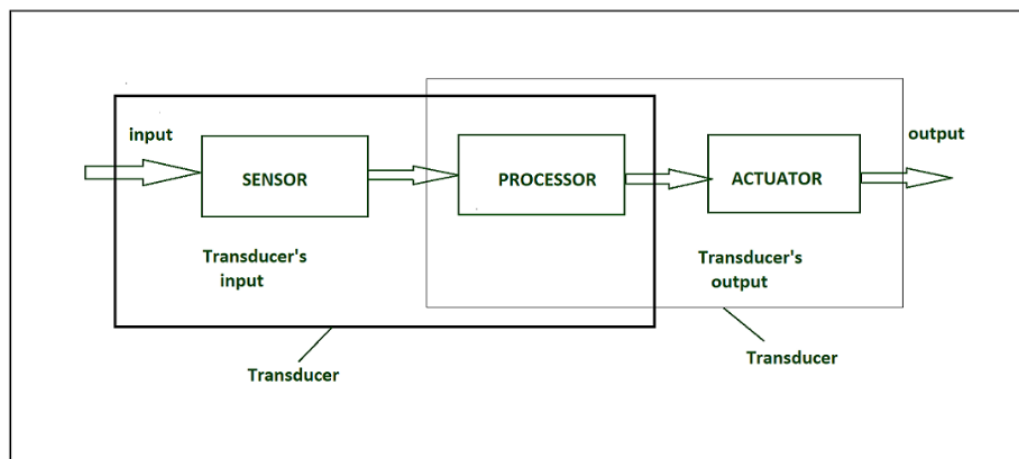
Sensors are used for sensing things and devices etc.

Definition

A device that provides a usable output in response to a specified measurement.

The sensor attains a physical parameter and converts it into a signal suitable for processing (e.g. electrical, mechanical, optical) the characteristics of any device or material to detect the presence of a particular physical quantity.

The output of the sensor is a signal which is converted to a human-readable form like changes in characteristics, changes in resistance, capacitance, impedance, etc.



IOT HARDWARE

Transducer :

A transducer converts a signal from one physical structure to another.

It converts one type of energy into another type.

It might be used as actuator in various systems.

Features of sensors:

Sensors play a crucial role in collecting data from the environment and converting it into electrical signals. Various sensor types exist, each designed to measure specific physical quantities.

Common features of sensors include:

1. sensitivity
2. Accuracy
3. precision.

Sensitivity: Sensitivity refers to a sensor's ability to detect subtle changes in the measured parameter.

Accuracy : Accuracy reflects how closely the sensor's readings align with the true values, while precision indicates the consistency of repeated measurements.

Additionally, sensors exhibit a dynamic range, representing the span between the smallest and largest detectable signals. Response time measures how quickly a sensor reacts to changes,

crucial in time-sensitive applications. Some sensors are designed for specific environments, offering robustness against factors like temperature, humidity, or electromagnetic interference.

Communication protocols enable sensors to transmit data to connected systems, often through wired or wireless channels. Power consumption is a critical consideration, influencing a sensor's lifespan and practicality in energy-efficient applications. Sensor fusion, combining data from multiple sensors, enhances overall system accuracy. As technology advances, miniaturization and integration enable sensors to become more compact, cost-effective, and capable of diverse applications across industries like healthcare, automotive, and IoT.

Resolution concept in iot technology:

In IoT (Internet of Things) technology, resolution refers to the level of detail or precision with which a sensor or device can capture and transmit data. It is a critical aspect that influences the quality and accuracy of information collected from the physical world.

Sensor resolution is commonly associated with the smallest detectable change in the measured parameter. For example, in a temperature sensor, resolution might determine the smallest temperature difference the sensor can reliably detect. Higher resolution enables the sensor to capture more fine-grained data, providing a more nuanced understanding of the environment.

Resolution is particularly crucial when dealing with analog signals, where it represents the smallest analog step that can be distinguished. In digital sensors, resolution often corresponds to the number of bits in the output data, influencing the precision of the measurements.

IoT applications benefit from high-resolution sensors as they can provide more detailed insights, leading to better-informed decision-making. However, increased resolution can also demand more resources, such as increased data storage and higher communication bandwidth. Therefore, achieving a balance between resolution and practical considerations is essential in designing effective and efficient IoT systems.

A good sensor should have these three characteristics:

1. It needs to be attentive to the phenomenon it's tracking.
2. It shouldn't be affected by other bodily factors.
3. During the measuring procedure, it should not change the phenomenon being measured.

We can use a variety of sensors to measure practically all of the physical properties around us. Thermometers, pressure sensors, light sensors, accelerometers, gyroscopes, motion sensors, gas sensors, and many other common sensors are widely used in everyday life. Several properties can be used to describe a sensor, the most essential of which is

Range - The sensor's range refers to the phenomenon's highest and minimum values.

Sensitivity Sensitivity is defined as the smallest change in the measured parameter that results in a noticeable change in the output signal.

Resolution - The sensor's resolution is the smallest change in the phenomenon it can detect.

Sensors characteristics :

Static

Dynamic

1. Static characteristics :

It is about how the output of a sensor changes in response to an input change after steady state condition.

Accuracy: Accuracy is the capability of measuring instruments to give a result close to the true value of the measured quantity. It measures errors. It is measured by absolute and relative errors. Express the correctness of the output compared to a higher prior system. Absolute error = Measured value – True value

Relative error = Measured value/True value

Range: Gives the highest and the lowest value of the physical quantity within which the sensor can actually sense. Beyond these values, there is no sense or no kind of response. e.g. RTD for measurement of temperature has a range of -200`c to 800`c.

Resolution: Resolution is an important specification for selection of sensors. The higher the resolution, better the precision. When the accretion is zero to, it is called the threshold. Provide the smallest changes in the input that a sensor is able to sense.

Precision: It is the capacity of a measuring instrument to give the same reading when repetitively measuring the same quantity under the same prescribed conditions.

It implies agreement between successive readings, NOT closeness to the true value.

It is related to the variance of a set of measurements.

It is a necessary but not sufficient condition for accuracy.

Sensitivity: Sensitivity indicates the ratio of incremental change in the response of the system with respect to incremental change in input parameters. It can be found from the slope of the output characteristics curve of a sensor. It is the smallest amount of difference in quantity that will change the instrument's reading.

Linearity: The deviation of the sensor value curve from a particularly straight line. Linearity is determined by the calibration curve. The static calibration curve plots the output amplitude versus the input amplitude under static conditions.

A curve's slope resemblance to a straight line describes linearity.

Drift: The difference in the measurement of the sensor from a specific reading when kept at that value for a long period of time.

Repeatability: The deviation between measurements in a sequence under the same conditions. The measurements have to be made under a short enough time duration so as not to allow significant long-term drift.

Dynamic Characteristics :

Properties of the systems

Zero-order system: The output shows a response to the input signal with no delay. It does not include energy-storing elements.

Ex. potentiometer measure, linear and rotary displacements.

First-order system: When the output approaches its final value gradually.

Consists of an energy storage and dissipation element.

Second-order system: Complex output response. The output response of the sensor oscillates before steady state.

Sensor Classification :

1.Passive & Active

2.Analog & digital

3.Scalar & vector

Passive Sensor –

Can not independently sense the input. Ex- Accelerometer, soil moisture, water level and temperature sensors.

Active Sensor –

Independently sense the input. Example- Radar, sonar and laser altimeter sensors.

Analog Sensor –

The response or output of the sensor is some continuous function of its input parameter. Ex- Temperature sensor, LDR, analog pressure sensor and analog hall effect.

Digital sensor –

Response in binary nature. Design to overcome the disadvantages of analog sensors. Along with the analog sensor, it also comprises extra electronics for bit conversion. Example – Passive infrared (PIR) sensor and digital temperature sensor(DS1620).

Scalar sensor –

Detects the input parameter only based on its magnitude. The answer for the sensor is a function of magnitude of some input parameter. Not affected by the direction of input parameters.

Example – temperature, gas, strain, color and smoke sensor.

Vector sensor –

The response of the sensor depends on the magnitude of the direction and orientation of input parameter. Example – Accelerometer, gyroscope, magnetic field and motion detector sensors.

Types of sensors –

Electrical sensor :

Electrical proximity sensors may be contact or non contact.

Simple contact sensors operate by making the sensor and the component complete an electrical circuit.

Non- contact electrical proximity sensors rely on the electrical principles of either induction for detecting metals or capacitance for detecting non metals as well.

Light sensor:

Light sensor is also known as photo sensors and one of the important sensor.

Light dependent resistor or LDR is a simple light sensor available today.

The property of LDR is that its resistance is inversely proportional to the intensity of the ambient light i.e when the intensity of light increases, it's resistance decreases and vice versa.

Touch sensor:

Detection of something like a touch of finger or a stylus is known as touch sensor.

It's name suggests that detection of something.

They are classified into two types:

Resistive type

Capacitive type

Today almost all modern touch sensors are of capacitive types.

Because they are more accurate and have better signal to noise ratio.

Range sensing:

Range sensing concerns detecting how near or far a component is from the sensing position, although they can also be used as proximity sensors.

Distance or range sensors use non-contact analog techniques. Short range sensing, between a few millimetres and a few hundred millimetres is carried out using electrical capacitance, inductance and magnetic technique.

Longer range sensing is carried out using transmitted energy waves of various types eg radio waves, sound waves and lasers.

Mechanical sensor:

Any suitable mechanical / electrical switch may be adopted but because a certain amount of force is required to operate a mechanical switch it is common to use micro-switches.

Pneumatic sensor:

These proximity sensors operate by breaking or disturbing an air flow.

The pneumatic proximity sensor is an example of a contact type sensor. These cannot be used where light components may be blown away.

Optical sensor:

In their simplest form, optical proximity sensors operate by breaking a light beam which falls onto a light sensitive device such as a photocell. These are examples of non contact sensors. Care must be exercised with the lighting environment of these sensors for example optical sensors can be blinded by flashes from arc welding processes, airborne dust and smoke clouds may impede light transmission etc.

Speed Sensor:

Sensor used for detecting the speed of any object or vehicle which is in motion is known as speed sensor. For example – Wind Speed Sensors, Speedometer, UDAR, Ground Speed Radar.

Temperature Sensor:

Devices which monitor and track the temperature and give temperature's measurement as an electrical signal are termed as temperature sensors. These electrical signals will be in the form of voltage and is directly proportional to the temperature measurement.

PIR Sensor:

PIR stands for passive infrared sensor and it is an electronic sensor that is used for the tracking and measurement of infrared (IR) light radiating from objects in its field of view and is also known as Pyroelectric sensor. It is mainly used for detecting human motion and movement detection.

Ultrasonic Sensor:

The principle of ultrasonic sensor is similar to the working principle of SONAR or RADAR in which the interpretation of echoes from radio or sound waves to evaluate the attributes of a target by generating the high frequency sound waves.

Actuation

A physical object ("thing") + controller ("brain") + sensors + actuators + networks make up an IoT device (Internet). A machine component or system that moves or regulates a mechanism or system is known as an actuator. The device's sensors detect the surroundings, and control signals are sent to the actuators based on the activities required.

An actuator is something like a servo motor. They can move to a defined angular or linear location and are either linear or rotatory actuators. For IoT applications, we can use servo motors and rotate them to 90 degrees, 180 degrees, and other angles as needed.

Through the actuator, the control system affects the environment. It necessitates an energy supply as well as a control signal. It turns the source of energy into a mechanical operation when it gets a control signal. On this premise, on the form of energy it uses, it has different types given below.

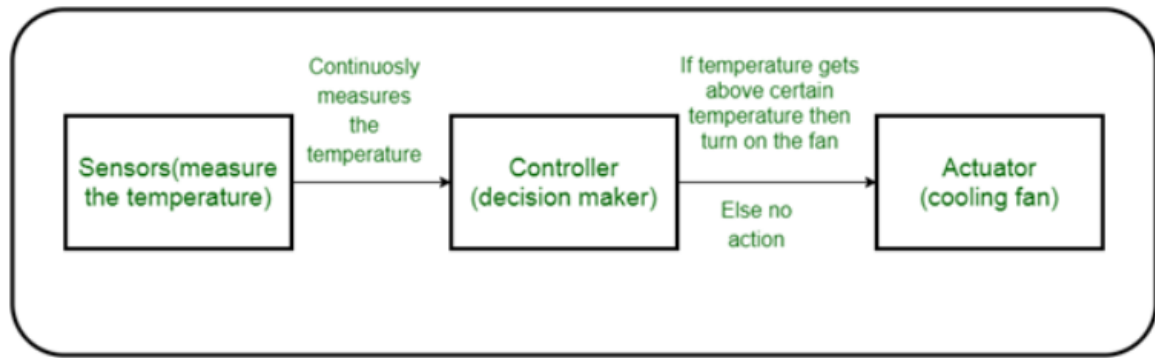


Fig 4: Use of actuations

Types of Actuators:

1. Hydraulic actuator:

A hydraulic actuator is a mechanical device that employs hydraulic power to complete a task. A cylinder or a fluid motor drives them. According to the needs of the lot device, mechanical motion is translated to rotary, linear, or oscillatory motion. Hydraulic actuators are used in construction equipment because they can create a considerable amount of force.

Advantages:

Hydraulic actuators have the ability to generate significant amounts of force at a high rate.

Used in welding, clamping, and other applications.

In car transport carriers, it's used to lower or raise the vehicles.

Disadvantages

Leaks in hydraulic fluid can reduce performance and complicate cleanup.

Noise reduction equipment, heat exchangers, and high-maintenance systems are all required.

It is expensive.

2. Pneumatic Actuators:

A pneumatic actuator converts energy created by vacuum or high-pressure compressed air into linear or rotary motion. For example, sensors that act like human fingers and are powered by compressed air are used in robotics

Advantages:

They are a low-cost solution that is employed in extreme temperatures where employing air rather than chemicals is a safer option.

They require little maintenance, are long-lasting, and have a lengthy service life.

It is quite quick to initiate and stop the action.

Disadvantages:

It can become less efficient if there is a loss of pressure.

The air compressor should be turned on all the time.

It is possible for air to be polluted, and it must be maintained.

3. Electrical actuators:

An electric actuator works by converting electrical energy into mechanical torque and is usually powered by a motor. A solenoid-based electric bell is an example of an electric actuator.

Advantages:

It can automate industrial valves, which makes it useful in a variety of sectors.

It makes less noise and is completely safe to use because there are no fluid leaks.

It has the ability to be reprogrammed and delivers the highest level of control and precision positioning.

Disadvantages:

It's not cheap.

It is highly dependent on the surrounding environment.

Purpose of Sensors and Actuators in IoT:

In IoT (Internet of Things), sensors and actuators play crucial roles. Sensors gather data from the physical world, such as temperature, light, or motion. This data is then sent to the IoT

system for analysis. Actuators, on the other hand, enable the IoT system to affect the physical world by triggering actions based on the analyzed data. Together, sensors and actuators facilitate the collection of real-world information and the execution of actions, forming the foundation for smart and responsive IoT applications.

In the realm of IoT, sensors and actuators are pivotal components that enable the digital integration of the physical world. Sensors serve as the eyes and ears of IoT devices, capturing diverse data from the environment. Whether measuring temperature, humidity, light, or motion, sensors translate real-world phenomena into digital information. This data is then transmitted to IoT platforms for analysis and decision-making.

Actuators, in turn, bridge the gap between the virtual and physical realms. These devices respond to commands from the IoT system by initiating actions in the physical world. For instance, actuators could control smart thermostats, turn on lights, or regulate machinery based on the insights derived from sensor data. Together, sensors and actuators create a dynamic feedback loop within the IoT ecosystem. Sensors continuously feed data for analysis, and actuators execute responses, making IoT systems agile, adaptive, and capable of influencing and optimizing the physical environment. This symbiotic relationship empowers IoT applications across various domains, from smart homes and cities to industrial automation, enhancing efficiency, sustainability, and user experience.

RFID Technology:

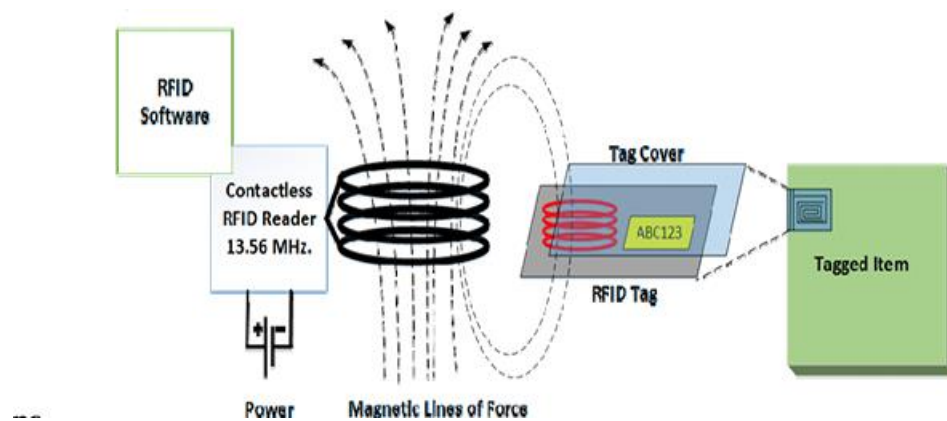
- RFID is an acronym for “radio-frequency identification”.
- Data digitally encoded in RFID tags, which can be read by a reader.
- Somewhat similar to barcodes.
- Data read from tags are stored in a database by the reader.
- As compared to traditional barcodes and QR codes, RFID tag data can be read outside the line-of-sight.

RFID Features

- RFID tag consists of an integrated circuit and an antenna.
- The tag is covered by a protective material which also acts as a shield against various environmental effects.
- Tags may be passive or active.
- Passive RFID tags are the most widely used.
- Passive tags have to be powered by a reader inductively before they can transmit information, whereas active tags have their own power supply.

Working Principle

- Derived from Automatic Identification and Data Capture (AIDC) technology.
- AIDC performs object identification, object data collection and mapping of the collected data to computer systems with little or no human intervention.
- AIDC uses wired communication.
- RFID uses radio waves to perform AIDC functions.
- The main components of an RFID system include an RFID tag or smart label, an RFID reader, and an antenna.



Applications

Inventory management 2. Asset tracking 3. Personnel tracking 4. Controlling access to restricted areas 5. ID badging 6. Supply chain management 7. Counterfeit prevention (e.g. in the pharmaceutical industry)

Connectivity Technologies

Communication Protocols: The following communication protocols have immediate importance to consumer and industrial IoTs:

- IEEE 802.15.4
- Zigbee
- 6LoWPAN
- Wireless HART
- Z-Wave
- ISA 100
- Bluetooth
- NFC
- RFID

Features of IEEE 802.15.4:

IEEE 802.15.4

- Well-known standard for low data-rate WPAN.
- Developed for low-data-rate monitoring and control applications and extended-life low-power-consumption uses.
- This standard uses only the first two layers (PHY, MAC) plus the logical link control (LLC) and service specific convergence sub-layer (SSCS) additions to communicate with all upper layers.
- Uses direct sequence spread spectrum (DSSS) modulation.
- Highly tolerant of noise and interference and offers link reliability improvement mechanisms.
- Low-speed versions use Binary Phase Shift Keying (BPSK).
- High data-rate versions use offset-quadrature phase-shift keying (O-QPSK).
- Uses carrier sense multiple access with collision avoidance (CSMA-CA) for channel access.
- Multiplexing allows multiple users or nodes interference-free access to the same channel at different times.
- Networking topologies defined are -- Star, and Mesh.

IEEE 802.15.4 supports two types of network node:

1. Full Function Device (FFD)
 - Can talk to all types of devices.
 - Supports full protocol.
2. Reduced Function Device (RFD)
 - Can only talk to an FFD.

- Lower power consumption.
- Minimal CPU/RAM required.

IEEE 802.15.4 Types:

1. Beacon Enabled Networks

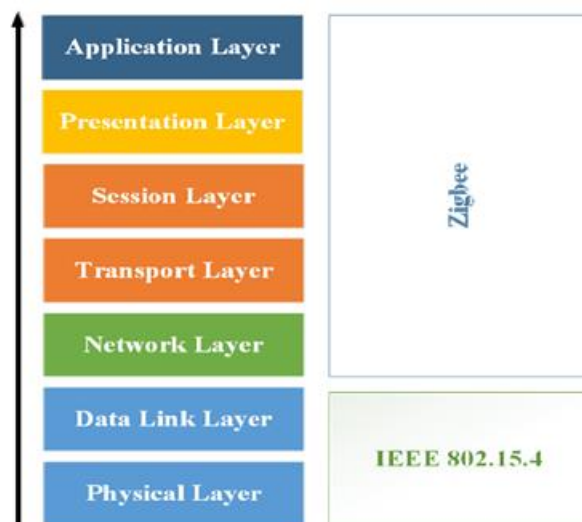
- Periodic transmission of beacon messages.
- Data-frames sent via Slotted CSMA/CA with a super frame structure managed by PAN coordinator. Beacons used for synchronization & association of other nodes with the coordinator.
- Scope of operation spans the whole network.

2. Non-Beacon Enabled Networks

- Data-frames sent via un-slotted CSMA/CA (Contention Based).
- Beacons used only for link layer discovery.
- Requires both source and destination IDs.
- As 802.15.4 is primarily, a mesh protocol, all protocol addressing must adhere to mesh configurations.
- De-centralized communication amongst nodes.

ZigBee

Features of ZigBee



- Most widely deployed enhancement of IEEE 802.15.4.
 - The ZigBee protocol is defined by layer 3 and above. It works with the 802.15.4 layers 1 and 2.
 - The standard uses layers 3 and 4 to define additional communication enhancements.
 - These enhancements include authentication with valid nodes, encryption for security, and a data routing and forwarding capability that enables mesh networking.
 - The most popular use of ZigBee is wireless sensor networks using the mesh topology.
- ZigBee has two important components:
- ZigBee Device Object (ZDO): ZDO responsible for Device management, Security, Policies.
 - Application Support Sub-layer (APS): APS responsible for Interfacing and control services, bridge between network and other layers

ZigBee Types

1. ZigBee Coordinator (ZC):

- The coordinator forms the root of the ZigBee network tree and might act as a bridge between networks.

- There is a single ZigBee Coordinator in each network, which originally initiates the network.

- It stores information about the network under it and outside it.

- It acts as a Trust Centre & repository for security keys.

2. ZigBee Router (ZR): Capable of running applications, as well as relaying information between nodes connected to it.

3. ZigBee End Device (ZED):

- It contains just enough functionality to talk to the parent node, and it cannot relay data from other devices.

- This allows the node to be asleep a significant amount of the time thereby enhancing battery life.

- Memory requirements and cost of ZEDs are quite low, as compared to ZR or ZC.

Applications:

- Building automation

- Remote control (RF4CE or RF for consumer electronics)

- Smart energy for home energy monitoring

- Health care for medical and fitness monitoring

- Home automation for control of smart homes

- Light Link for control of LED lighting

- Telecom services.

6LoWPAN

- Low-power Wireless Personal Area Networks over IPv6.

- Allows for the smallest devices with limited processing ability to transmit information wirelessly using an Internet protocol.

- Allows low-power devices to connect to the Internet.

- Created by the Internet Engineering Task Force (IETF) - RFC 5933 and RFC 4919.

Features of 6LoWPANs

- Allows IEEE 802.15.4 radios to carry 128-bit addresses of Internet Protocol version 6 (IPv6).

- Header compression and address translation techniques allow the IEEE 802.15.4 radios to access the Internet.

- IPv6 packets compressed and reformatted to fit the IEEE 802.15.4 packet format.

- Uses include IoT, Smart grid, and M2M applications.

Addressing in 6LoWPAN

- 64-bit addresses: globally unique.

- 16-bit addresses: PAN specific; assigned by PAN coordinator

6LoWPAN Routing

- Mesh routing within the PAN space.

- Routing between IPv6 and the PAN domain

- Routing protocols in use:

- LOADng

- RPL

LOADng Routing

Basic operations of LOADng include:

- Generation of Route Requests (RREQs) by a LOADng Router (originator) for discovering a route to a destination,

- Forwarding of such RREQs until they reach the destination LOADng Router,

- Generation of Route Replies (RREPs) upon receipt of an RREQ by the indicated destination, and unicast hop-by-hop forwarding of these RREPs towards the originator.

- If a route is detected to be broken, a Route Error (RERR) message is returned to the originator of that data packet to inform the originator about the route

breakage.

IP Addressing in IoT:

- Every IoT device must have a unique **IP address** to communicate over the Internet.
- **IPv6** is preferred due to its vast address space, essential for billions of devices.
- **6LoWPAN** enables IPv6 addressing on low-power devices.
- Address types: **Global, Local, Link-local, and Unique local addresses.**
- Supports both static and dynamic addressing methods.

Medium Access Control (MAC):

- The MAC layer controls how devices access the communication channel.
- **CSMA/CA** (Carrier Sense Multiple Access with Collision Avoidance) is widely used.
- Responsibilities:
 - Avoiding data collisions.
 - Managing retransmissions and acknowledgments.
 - Ensuring fair channel access.
- Used in IEEE 802.15.4, Wi-Fi, and Bluetooth protocols.

Application Layer Protocols in IoT:

- These protocols define how IoT applications exchange data.
- Common IoT Application Layer Protocols:
 - **MQTT** – Lightweight publish/subscribe messaging protocol.
 - **CoAP** – Designed for constrained devices, uses REST model.
 - **HTTP/HTTPS** – Common for web-based IoT applications.
 - **AMQP** – Reliable message delivery protocol for enterprise use.
 - **DDS** – Real-time data distribution in critical systems.
- Role:
 - Facilitate structured data communication.
 - Ensure secure, scalable, and reliable interactions.